

Total No. of Questions : 4]

SEAT No. :

PF196

[Total No. of Pages : 2

APR-26/SE/Insem-242

S.E. (Civil) (Insem)

SURVEY

(2019 Pattern) (Semester - IV) (201009)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Solve Q.1 or Q.2 and Q.3 or Q.4.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data if necessary.*
- 5) *Use of electronic pocket calculator is allowed.*
- 6) *Use of cell phone is prohibited in examination hall.*

Q1) a) Explain the principle of surveying. **[4]**

b) Explain intersection method of plane table surveying with neat sketch. **[4]**

c) The following consecutive readings were taken with an auto level and a 4-meter levelling staff on a continuously sloping ground at common intervals of 30 m.

0.855 (on A), 1.545, 2.335, 3.115, 3.825, 0.455, 1.380, 2.055, 2.855, 3.455, 0.585, 1.015, 1.850, 2.755 and 3.845 (on B). **[7]**

The RL of A was 380.500. Prepare a tabular form and calculate the RLs of different points using Rise and Fall method. Apply the usual checks. Also, determine the gradient of AB.

OR

Q2) a) Explain in detail the reciprocal type of leveling. **[4]**

b) Explain the following terms: **[4]**

- i) Meridian
- ii) Declination
- iii) Benchmark
- iv) Ranging

P.T.O.

- c) The following are the observed bearings of the lines of a traverse ABCDEA with a compass in a place where local attraction was suspected. Find the correct bearings of the lines. [7]

Line	FB	BB
AB	191°45'	13°0'
BC	39°30'	222°30'
CD	22°15'	200°30'
DE	242°45'	62°45'
EA	330°15'	147°45'

- Q3) a) Write a short note on Prolongation of straight-line using theodolite. [4]
 b) State and explain the fundamental axes of theodolite. [5]
 c) Was not recorded. Compute the length and bearing of the line DA. [6]

Line	Length (m)	Bearing
AB	75.50	30°24'
BC	180.50	110°36'
CD	60.25	210°30'
DA	?	?

OR

- Q4) a) Explain the following terms: [4]
 i) Telescope inverted
 ii) Swinging of telescope
 iii) Latitude
 iv) Departure
 b) Enlist and explain the permanent adjustment of transit theodolite. [5]
 c) Determine the missing data for the following table of a closed traverse ABCDA. [6]

Line	AB	BC	CD	DA
Length (m)	230.50	250.20	?	?
Bearing	N36°45'E	S82°48'E	S10°15'E	N82°43'W

